

moustaches

Fench statistics (tables and diagrams)

v0.1.0

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MIT

About

While preparing my lessons, I realized that none of the existing packages for creating statistical diagrams support counts, the French definitions of quartiles, or the construction of histograms and box plots as I teach them... This package was created to fill that gap.

I took inspiration from parts of the LaTeX package ProfCollege (by Christophe Poulain), but the implementation is not a copy, and I did not use AI at all in the development of this package.

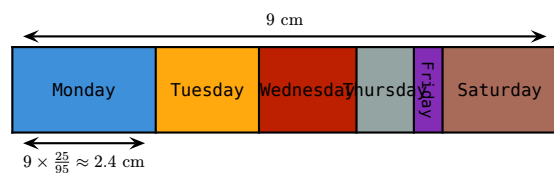
Main module: “stat”

- `bandes()`
- `camembert()`
- `caracteristiques()`
- `histogramme()`
- `pgcd()`
- `simpli()`
- `stat()`

bandes

Draws a band chart.

```
#bandes(  
  valeurs:("Monday", "Tuesday",  
    "Wednesday", "Thursday", "Friday",  
    "Saturday"),  
  effectifs:(25, 18, 17, 10, 5, 20),  
  width:9cm,  
  tourne:(4,),  
  explication:(0,),  
)
```



Parameters

```
bandes(  
  valeurs: array ,  
  effectifs: array ,  
  width: length ,  
  tourne: array ,  
  couleurs: auto array ,  
  print: boolean ,  
  explication: array ,  
  hauteur: length  
) -> content
```

valeurs array

Categories (modalities)

Default: ()

effectifs array

Counts of each category

Default: ()

width length

Total width of the chart

Default: 1fr

tourne array

Indices of labels to rotate by 90°

Default: ()

couleurs auto or array

List of rectangle colors

Default: auto

print boolean

If true, replaces colors with tiling patterns

Default: false

explication array

Construction steps to explain, e.g. (0,) or (0,3)

Default: ()

hauteur length

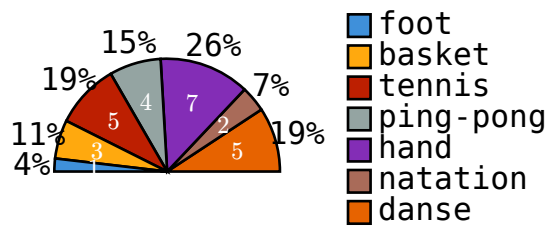
Height of the band chart

Default: 40pt

camembert

Draws a (semi-)pie chart.

```
#camembert(  
  valeurs:  
    ("foot", "basket", "tennis", "ping-  
    pong", "hand", "natation", "danse"),  
  effectifs:(1, 3, 5, 4, 7, 2, 5),  
  radius:1.5,  
  legende:"d",  
  espace:0,  
  semi:true,  
)
```



Parameters

```
camembert(  
  valeurs: array,  
  effectifs: array,  
  data: array,  
  semi: boolean,  
  legende: str,  
  espace: int float,  
  print: boolean,  
  couleurs: auto array,  
  dedans: boolean none dictionary,  
  dehors: boolean none dictionary,  
  ..args  
) -> content
```

valeurs array

Categories (modalities)

Default: ()

effectifs array

Counts of each category

Default: ()

data array

Allows passing a single data array instead

Default: ()

semi `boolean`

If `true`, draws a semicircular chart

Default: `false`

legende `str`

Legend position: "d" (right), "g" (left), "h" (top), "b" (bottom), or "" for none

Default: `"d"`

espace `int` or `float`

Spacing between legend items when the legend is on the left or right

Default: `.1`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

couleurs `auto` or `array`

List of sector colors

Default: `auto`

dedans `boolean` or `none` or `dictionary`

`true`, `none`, `false`, or options passed to inner-label

Default: `true`

dehors `boolean` or `none` or `dictionary`

`true`, `none`, `false`, or options passed to outer-label

Default: `true`

caracteristiques

Computes descriptive statistics and returns them as a dictionary: effectif-total, mediane, classe-mediane, q1, q3, d1, d9, ecart-interquartiles, max, min, etendue, modes, effectif-modes, sommex, sommex2, moyenne, variance, ecart-type, variance-echantillon, ecart-type-echantillon, lqboxplot, and cetzboxplot.

```
#caracteristiques(sondage:(1, 5, 7, 9,  
4, 5, 9, 8, 7, 3, 8))
```

```
(  
  effectif-total: 11,  
  mediane: 7,  
  classe-médiane: none,  
  q1: 4,  
  q3: 8,  
  d1: 3,  
  d9: 9,  
  ecart-interquartiles: 4,  
  max: 9,  
  min: 1,  
  etendue: 8,  
  modes: (5, 7, 8, 9),  
  effectif-modes: 2,  
  sommex: 66,  
  sommex2: 464,  
  moyenne: 6.0,  
  variance: 6.1818181818181818,  
  ecart-type: 2.4863262420322436,  
  variance-echantillon: 6.7999999999999997,  
  ecart-type-echantillon: 2.607680962081059,  
  lqboxplot: (  
    median: 7,  
    q1: 4,  
    q3: 8,  
    whisker-low: 3,  
    whisker-high: 9,  
    outliers: (1,),  
    mean: 6.0,  
  ),  
  cetzboxplot: (  
    q1: 4,  
    q2: 7,  
    q3: 8,  
    min: 3,  
    max: 9,  
    outliers: (1, 6.0),  
  ),  
)
```

Parameters

```
caracteristiques(  
  valeurs: array,  
  effectifs: array,  
  classes: array,  
  bins: int,  
  sondage: array,  
  crochets: boolean,  
  deciles: boolean  
) -> dictionary
```

valeurs array

Values of the numeric variable

Default: ()

effectifs array

Counts of each value

Default: ()

classes array

Class boundaries, if applicable: `length = effectifs.len() + 1`

Default: ()

bins int

For raw data in `sondage`, automatically creates bins classes of equal width

Default: 0

sondage array

List of all observations

Default: ()

crochets boolean

For class intervals, use interval notation (`[a; b[`) instead of inequalities (`a <= ... < b`)

Default: `false`

deciles boolean

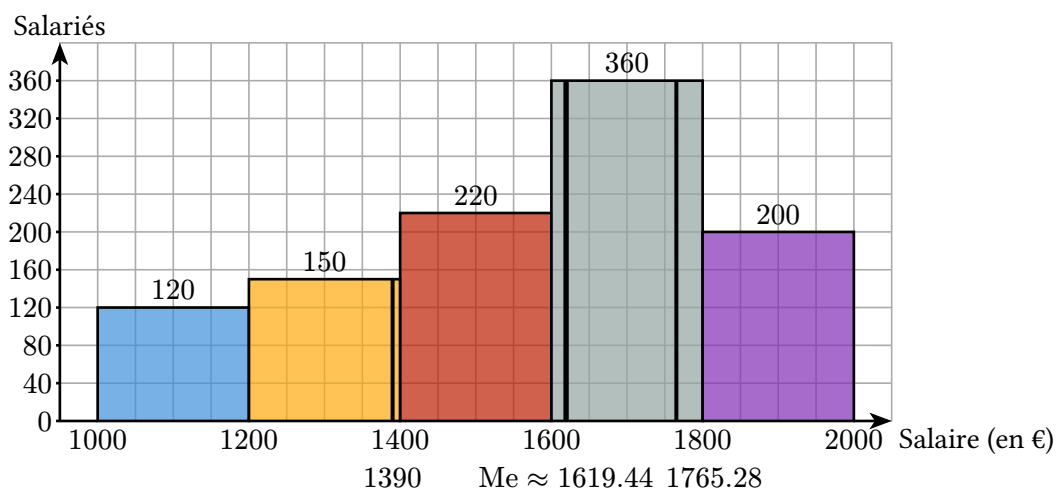
If `true`, whiskers are based on deciles; otherwise they extend to $\pm 1.5 \times \text{IQR}$

Default: `true`

histogramme

Draws a true histogram: class widths do not have to be equal, and rectangle areas are proportional to the counts/frequencies.

```
#histogramme(
  classes: (1000, 1200, 1400, 1600, 1800, 2000),
  effectifs: (120, 150, 220, 360, 200),
  px: 50,
  uaire: 10,
  s: .5,
  // fontsize: .5em,
  nom-donnee: [Salaire (en €)],
  nom-effectifs: [Salariés],
  mediane: true,
  quartiles: true,
  pos-rect: (0,5),
)
```



Parameters

```
histogramme(  
    classes: array ,  
    effectifs: array ,  
    px: int ,  
    uaire: int ,  
    s: int float dictionary ,  
    grille: boolean ,  
    pas-grille: int float ,  
    nom-donnee: content ,  
    nom-effectifs: content ,  
    add: int array ,  
    lecture: boolean ,  
    donnees-sup: boolean ,  
    liste-couleurs: auto color array ,  
    pos-rect: array ,  
    print: boolean ,  
    transparence: ratio ,  
    ecc: boolean ,  
    mediane: boolean ,  
    quartiles: boolean ,  
    decimales: int ,  
    posq1: array ,  
    posmediane: array ,  
    posq3: array ,  
    stroke-mediane: length stroke ,  
    stroke-quartiles: length stroke ,  
    textsize: length  
) -> content
```

classes array

Class boundaries: $\text{length} = \text{effectifs}.\text{len}() + 1$

Default: ()

effectifs array

Counts of each class: $\text{length} = \text{classes}.\text{len}() - 1$

Default: ()

px int

Tick interval on the x-axis

Default: 1

uaire `int`

Quantity represented by one grid square

Default: `1`

s `int` or `float` or `dictionary`

Scale passed to cetz

Default: `1`

grille `boolean`

Whether to display a grid

Default: `true`

pas-grille `int` or `float`

Grid spacing

Default: `1`

nom-donnee `content`

Label for the legend or y-axis, if applicable

Default: `[Values]`

nom-effectifs `content`

Label for the x-axis

Default: `[Frequencies]`

add `int` or `array`

Extra margin on the left, right, and bottom. Either a single integer or a triple

Default: `(-1, 1, 1)`

lecture `boolean`

Whether to display a y-axis when all classes have the same width

Default: `true`

donnees-sup `boolean`

Whether to display frequencies above the rectangles

Default: `true`

liste-couleurs `auto` or `color` or `array`

Rectangle colors, or a single color (e.g. white for printing)

Default: `auto`

pos-rect `array`

Position of the legend rectangle relative to the upper-left corner of the smallest rectangle

Default: `(0, 2)`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

transparence `ratio`

Rectangle transparency. Use 100% if `ListeCouleurs: white`

Default: `30%`

ecc `boolean`

Displays cumulative frequencies

Default: `false`

mediane `boolean`

Displays the median

Default: `false`

quartiles `boolean`

Displays the quartiles

Default: `false`

decimales `int`

Number of decimal places for rounded values

Default: `2`

posq1 `array`

Position of the Q1 label relative to the x-axis

Default: `(0, -1.5)`

posmediane `array`

Position of the median label relative to the x-axis

Default: `(0, -1.5)`

posq3 `array`

Position of the Q3 label relative to the x-axis

Default: `(1, -1.5)`

stroke-mediane `length` or `stroke`

Stroke used to draw the median

Default: `2pt`

stroke-quartiles `length` or `stroke`

Stroke used to draw the quartiles

Default: `1.5pt`

textsize `length`

Text size

Default: `1em`

pgcd

Computes the greatest common divisor (GCD) of a list of numbers.

```
#pgcd(1000, 1200, 1400, 1600, 1800,  
2000)
```

200

Parameters

```
pgcd(..a: array) -> int
```

..a array

List of numbers

simpli

Writes the quotient a / b , where a and b are floating-point numbers (or simply a if $b = 1$), together with its reduced fractional form (up to 6 decimal places).

```
#simpli(3.6,6.9)
```

$$\frac{3.6}{6.9} = \frac{12}{23}$$

Parameters

```
simpli(  
  a,  
  b  
) -> content
```

a

int | float

b

int | float

stat

Performs a statistical analysis and displays the results as a table, with optional frequencies (effectifs), relative frequencies ("f" for fractions, "d" for decimals, "t" for all, true for percentages), sector angles for (semi-)pie charts, cumulative frequencies (ECC, FCC), class intervals (automatically generated when $\text{bins} > 0$) and their centers.

It can also display charts: "hbar", "bar", "circ", "semicirc", "box", "bande", "histo", or any combination of these.

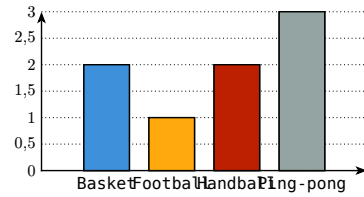
If `multi != ()`, grouped bar charts (horizontal or vertical) or multiple box plots can be displayed. Setting `print: true` replaces colors with tiling patterns.

```
#stat(
  classes:(150, 160, 170, 200),
  effectifs:(3, 4, 5),
  frequences:"f",
)
```

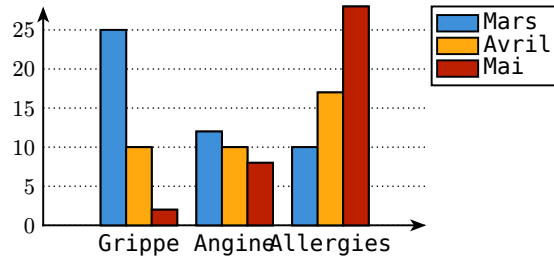
	150 ≤ ... < 160	160 ≤ ... < 170	170 ≤ ... < 200
Centres des classes	155	165	185
Effectifs	3	4	5
Fréquences	$\frac{3}{12} = \frac{1}{4}$	$\frac{4}{12} = \frac{1}{3}$	$\frac{5}{12}$

```
#stat(
  sondage:
    ("Handball", "Basket", "Football", "Handball",
     "Ping-pong", "Basket", "Ping-pong", "Ping-pong"),
  diagramme:"bar",
  cbar:(size:(8.2,4))
)
```

	Basket	Football	Handball	Ping-pong
Effectifs	2	1	2	3
Fréquences (en %)	25	13	25	38



```
#stat(
  valeurs:
    ("Grippe", "Angine", "Allergies"),
  multi:((25,12,10),(10,10,17),
         (2,8,28)),
  diagramme:"bar",
  labels:("Mars", "Avril", "Mai"),
  tableau:false
)
```



Parameters

```
stat(  
  valeurs: array,  
  effectifs: array,  
  qualitatif: boolean,  
  totaux: boolean,  
  tableau: boolean,  
  couleur-tableau: color,  
  inset: length,  
  nom-donnee: content,  
  nom-effectifs: content,  
  decimales: int,  
  frequences: boolean str,  
  sondage: array,  
  vide: boolean,  
  angle: boolean str,  
  ecc: boolean str,  
  classes: array,  
  bins: int,  
  crochets: boolean,  
  centre: boolean str,  
  colonnes-vide: array,  
  cases-vide: array,  
  diagramme: str,  
  bar: array,  
  cbar: array,  
  moustaches: array,  
  cmoustaches: array,  
  circ: dictionary,  
  bande: array,  
  histo: array,  
  fcc: boolean str,  
  multi: array,  
  labels: array,  
  print: boolean,  
  liste-couleurs: auto color array  
) -> content
```

valeurs array

Values or categories of the variable

Default: ()

effectifs array

Counts of each value

Default: ()

qualitatif `boolean`

Whether the variable is qualitative

Default: `true`

totaux `boolean`

Display totals in tables

Default: `false`

tableau `boolean`

Display the table

Default: `true`

couleur-tableau `color`

Background color of the first row and first column

Default: `luma(85%)`

inset `length`

Cell inset

Default: `5pt`

nom-donnee `content`

Label for the upper-left table cell (and some charts)

Default: `[]`

nom-effectifs `content`

Label for the frequency row (and some charts)

Default: `[Frequencies]`

decimales `int`

Number of decimal places for rounded values

Default: `0`

frequencies `boolean` or `str`

Relative frequencies: `true` for percentages, "`f`" for fractions, "`d`" for decimals, "`t`" for all, "`v`" for an empty row

Default: `true`

sondage `array`

List of all observations

Default: `()`

vide `boolean`

If `true`, all rows except the first are left blank

Default: `false`

angle `boolean` or `str`

Display sector angles ("`v`" for an empty row)

Default: `false`

ecc `boolean` or `str`

Display cumulative frequencies ("`v`" for an empty row)

Default: `false`

classes `array`

Class boundaries for continuous data: `length = effectifs.len() + 1`

Default: `()`

bins `int`

For raw data, automatically creates bins classes of equal width

Default: `0`

crochets `boolean`

Use interval notation (`[a; b[`) instead of inequalities (`a <= ... < b`)

Default: `false`

centre `boolean` or `str`

Display class centers ("v" for an empty row)

Default: `true`

colonnes-vide `array`

Columns to leave empty

Default: `()`

cases-vide `array`

Individual cells to leave empty, numbered row by row

Default: `()`

diagramme `str`

Chart type: "hbar", "bar", "circ", "semicirc", "box", "bande", "histo", or any combination

Default: `" "`

bar `array`

Parameters passed to the horizontal/vertical bar chart canvas

Default: `()`

cbar `array`

Parameters passed to cetz's barchart or columnchart

Default: `()`

moustaches `array`

Parameters passed to the custom box-plot function

Default: `()`

cmoustaches `array`

Parameters passed to cetz's boxwhiskers

Default: `()`

circ dictionary

Parameters passed to camembert

Default: (:)

bande array

Parameters passed to bandes

Default: ()

histo array

Parameters passed to histogramme

Default: ()

fcc boolean or str

Display cumulative relative frequencies ("v" for an empty row)

Default: false

multi array

Multiple series for grouped bar charts or multiple box plots

Default: ()

labels array

Labels for the different series

Default: ()

print boolean

If true, replaces colors with tiling patterns

Default: false

liste-couleurs auto or color or array

Colors used in the charts

Default: auto

Sub-module for tilings and colormaps: “hachures”

- `alveoles()`
- `cross()`

- dots()
- etoile()
- grille()
- nelines()
- nwlines()
- stripes()

Variables

- hachures
- typst18
- cycles-couleurs

alveoles

hexagons tiling

Parameters

```
alveoles(
  c: color,
  size: length,
  background: color,
  fill: none color
) -> tiling
```

cross

colored cross tiling that print in b&w

Parameters

```
cross(
  c: color,
  size: length,
  fill: auto color
) -> tiling
```

dots

colored dots tiling that print in b&w

Parameters

```
dots(
  c: color,
  size: length,
  fill: auto color
) -> tiling
```

etoile

stars tiling

Parameters

```
etoile(  
  c: color,  
  size: length,  
  background: color,  
  fill: none color  
) -> tiling
```

grille

grid lines tiling

Parameters

```
grille(  
  c: color,  
  size: length,  
  fill: color  
) -> tiling
```

nelines

North-east lines tiling

Parameters

```
nelines(  
  c: color,  
  size: length,  
  fill: color  
) -> tiling
```

nwlines

North-west lines tiling

Parameters

```
nwlines(  
  c: color,  
  size: length,  
  fill: color  
) -> tiling
```

stripes

colored stripes tiling that print in b&w

Parameters

```
stripes(  
  c: color,  
  size: length,  
  fill: auto color  
) -> tiling
```

hachures array

List of tilings

typst18 array

colormap of basic typst colors

cycles-couleurs array

25 cycles de couleurs disponibles